

Joint biclustering organizes region–feature co-variation across C1 and C2

Joint biclustering of the C1 and C2 count matrices revealed a strongly polarized organization of region–feature co-variation in the integrated dataset (Figure 1). Columns (assay–feature pairs) arrange along a pronounced left-to-right gradient of signal: early column groups are predominantly low, central groups are mixed, and late groups are predominantly high. This monotonic trend defines a latent activity axis across features, along which the row groups (genomic regions) are correspondingly stratified.

Row groups align with this axis, ranging from modules that are uniformly low across features (top) to modules that are strongly enriched in the rightmost, high-signal column blocks (bottom). Coherent bicluster structure dominates the heatmap as large contiguous blocks, indicating sets of regions and features that co-vary. Deviations are localized and interpretable: occasional narrow higher-than-baseline streaks within the leftmost low-signal column blocks highlight small feature subsets that depart from the global low baseline, whereas scattered lower-than-expected rows at the margins of the rightmost high-signal blocks mark region subsets that do not share the otherwise pervasive enrichment. Central column blocks interleave low and moderate signals, capturing intermediate modules and transitions between the extremes.

We refer to these bicluster-derived groups as modules: region modules (rows; labeled A–O in downstream figures) and feature modules (columns). These modules provide the units of aggregation for all subsequent analyses—enrichment testing, comparative summaries, and

annotation transfer across C1 and C2—ensuring that downstream interpretation operates on coherent, jointly defined patterns of co-variation established in the integrated dataset.

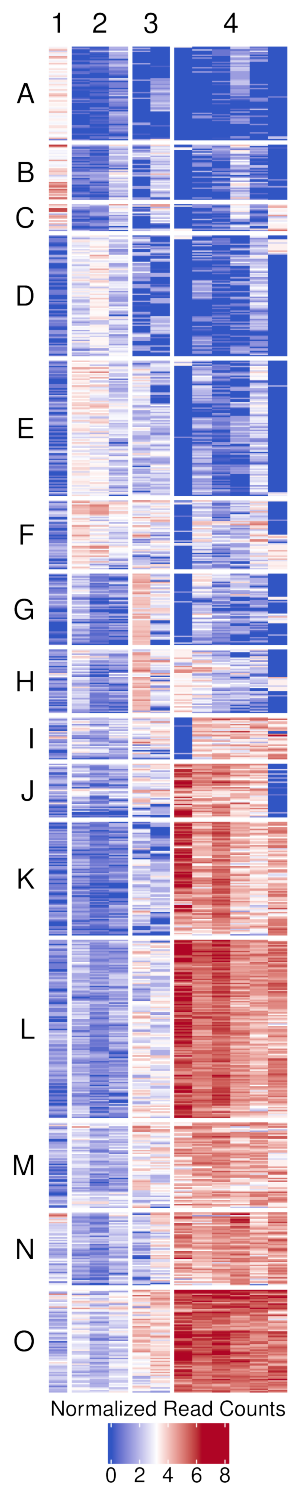


Figure 1. Biclustering highlights region–feature co-variation across C1 and C2. Rows (genomic regions) and columns (feature groups) are jointly clustered from the combined C1 and C2 count matrix, with colors indicating normalized signal. Left column clusters are largely low, while right clusters are high, forming coherent blocks that define sets of co-varying regions and features.

Transcription factor motif signals align with bicluster structure

Motif enrichment mirrors the biclustered region structure, revealing a pronounced early-to-late polarization in putative regulatory engagement (Figure 2). Using hg38 motif annotations, enrichment is broadly depleted in the leftmost region modules and elevated across later modules, indicating that most transcription factor (TF) recognition sequences are preferentially concentrated in later modules. This left–right gradient suggests a global shift in regulatory potential across the biclustered modules, with later modules harboring sequence or accessibility contexts more competent for TF binding.

Superimposed on this global elevation, a subset of motifs exhibits sharp, module-specific enrichment peaks that form vertical bands in particular right-hand modules (Figure 2). These peaks nominate the corresponding TFs (as annotated by their motifs) as candidate drivers of module-specific regulatory programs. In contrast, motifs that are uniformly elevated across most later modules likely reflect broadly acting factors—such as ubiquitous or pioneer-like regulators—providing a shared activation backdrop rather than module-restricted control.

Operationally, candidate regulators can be prioritized per module by selecting motifs with (i) high absolute enrichment in that module and (ii) strong specificity, for example a high ratio of within-module to median enrichment across modules. Sharply peaked motifs in specific late modules define focused hypotheses for module-restricted regulators, whereas uniformly elevated motifs in late modules indicate common upstream inputs. Integrating these motif candidates with orthogonal evidence—such as TF expression dynamics across the C1 and C2 contexts, motif deviation scores, or footprinting within the corresponding region sets—will refine regulator lists and resolve family-level motif redundancy. Collectively, the alignment of motif signals with the bicluster architecture supports a model in which early region modules are comparatively TF-poor, whereas later modules are

broadly TF-enriched with superimposed module-specific TFs that likely execute distinct regulatory programs.

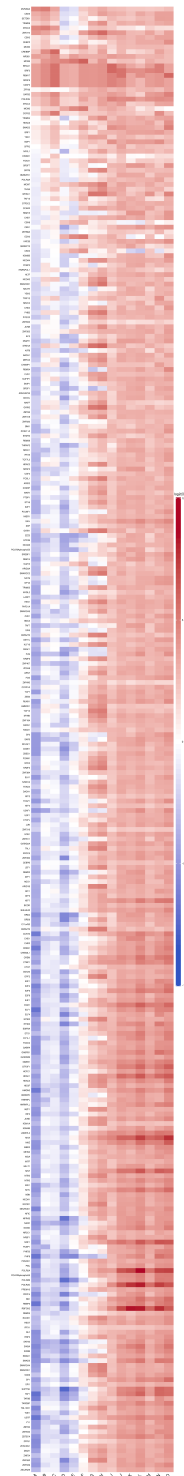


Figure 2. TF motif enrichment segregates C1–C2 region clusters. *Enrichment scores for transcription factor binding motifs are shown across region clusters from the joint C1–C2 biclustering. Most motifs are reduced in clusters 1–3 but elevated across clusters 4 onward, with a few motifs peaking in specific right-side clusters.*

Genomic and cCRE annotations reveal promoter-enhancer shifts across biclusters

Joint analysis of ENCODE cCRE classes and genic annotations demonstrated a coordinated shift from enhancer-dominated to promoter-enriched compositions across region modules A–O. In the cCRE space, early modules (A–B) displayed mixed categories with sizable “other” and CA/CTCF-associated components alongside enhancer-like elements (predominantly dELS) and limited promoter-like signatures (PLS). Intermediate modules (C–H) remained enhancer-skewed—most notably H, which was strongly dELS—with only small promoter or CA-related fractions. From I onward, PLS became the dominant class, increasing progressively through J–O, while “other” nearly disappeared and CA/CTCF/TF-associated categories persisted only as minor constituents (Figure 3). Parallel genic-annotation trends recapitulated this transition: early modules (A–D) were intergenic-dominated with intron and exon as secondary components and minimal promoter overlap; intermediate modules (E–H) retained notable intergenic content and higher intron fractions; beginning at I, promoter annotations became the largest component with a marked reduction in intergenic and intron fractions, and exon content increased concordantly—most prominently around I–K—while 5'/3' UTRs remained consistently minor across all modules (Figure 4). Together, the diminishing “other” classes, rising PLS, and concordant gains in promoter/exon with losses of intergenic/intron content indicate that the modules

order regulatory landscapes along a distal-enhancer-to-promoter axis, with later modules capturing promoter-proximal, gene-anchored regulatory architecture.

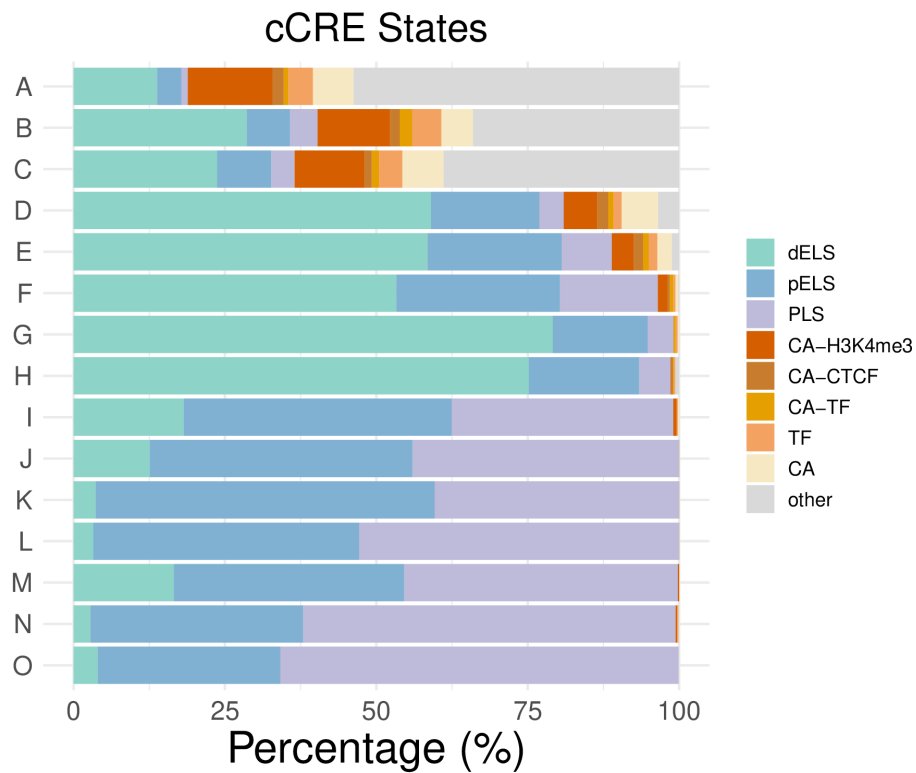


Figure 3. cCRE category composition across biclusters A–O. Biclustered *C1–C2* regions were mapped to ENCODE cCRE categories and summarized per group. Early groups (A–H) are dominated by enhancer-like states, especially dELS, with A–B also showing substantial “other” and CA-associated elements; later groups (I–O) are predominantly promoter-like (PLS) with minor pELS and small CA/CTCF/TF contributions.

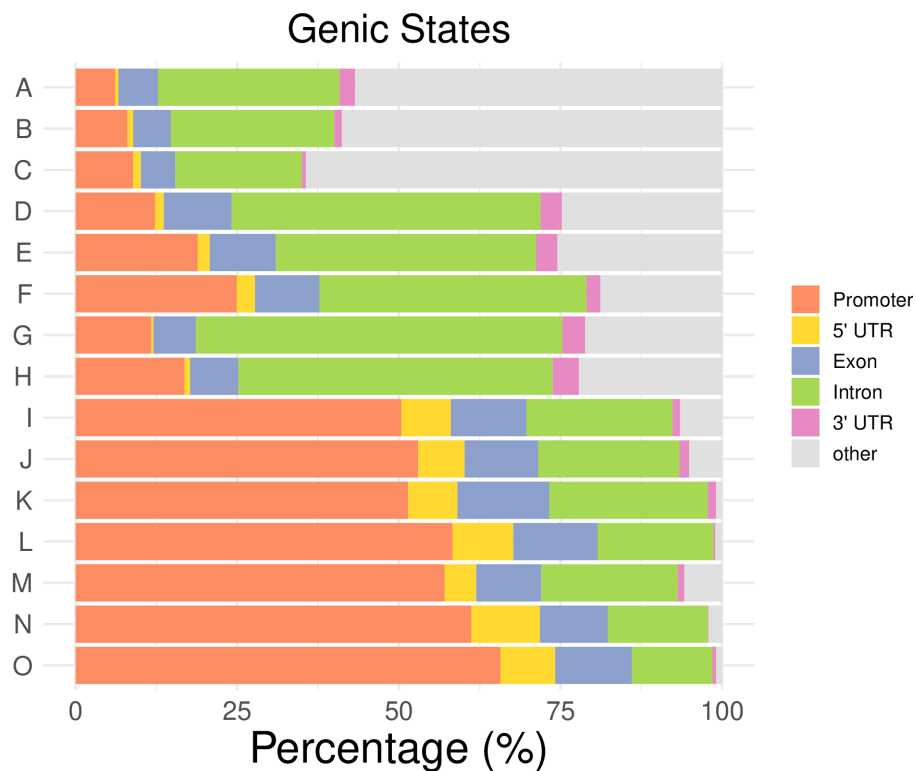


Figure 4. Genic annotations across bicluster groups show promoter shift. Biclustered regions are summarized by hg38 genic categories across groups A–O, showing a transition from intergenic/intronic compositions in early groups to promoter-enriched profiles in later groups. Exons are most prominent in mid-range groups (around I–K), while both 5' and 3' UTRs remain minor throughout.

References

1. Vanderstichele, T., Burnham, K. L., de Klein, N., Tardaguila, M., Howell, B., Walter, K., Kundu, K., Koeppel, J., Lee, W., Tokolyi, A., Persyn, E., Nath, A. P., Marten, J., Petrovski, S., Roberts, D. J., Di Angelantonio, E., Danesh, J., Berton, A., Platt, A., et al. (2024). Misexpression of inactive genes in whole blood is associated with nearby rare structural variants. *The American Journal of Human Genetics*, 111(8), 1524â 1543. <https://doi.org/10.1016/j.ajhg.2024.06.017>